

HW4: Individual Assignment

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I built a total of 2 nodes for this assignment: a Monte Carlo Localization (MCL) node and a localization error plotter. The MCL node applies the algorithm from the Probabilistic Robotics book. It uses the motion update from Table 5.6 and uses a likelihood field for the sensor update. The plotter node calculates the position and orientation error of the most likely pose compared to the ground truth from the simulation. I used the 2D pose estimation tool from RVIZ for the pose reset instead of a service. I found this to be faster than changing the pose for every service call.

At first, my localization was not working due to two challenges. The first challenge was testing in a hard map, where the general layout is symmetric. This symmetry means there are multiple positions that, from the robot's perspective look the same. This means that multiple distant particles can gain high weights and can converge to an incorrect pose. The second challenge was that my pose estimates generated particles with low variance, leading them to converge fast on wrong poses. To solve this, I increased the covariance of the initial pose estimates so that the particles could cover more area, and hopefully spawn more particles closer to the actual pose. This assignment showed me how challenging proper localization is, as it can be very dependent on initialization and can very easily converge to wrong estimates on hard symmetric maps.